Stanley Wang

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Education

McGill University, B.Sc. in Computer Science Sep 2022 – Dec 2026

Experience

Electronic Arts | Commerce & Identity May 2026 – Aug 2026
Software Engineering Intern – Python, Java, TypeScript, SQLite *Redwood City, CA*

- **Agentic Commerce**: Prototyped a shopping assistant that personalizes and stages Madden Mobile purchases
- Recorded purchases only after a service-authenticated callback matched the staged cart; made checkout retries idempotent
- Wrote an agent runtime with FastAPI and OpenAI Agents SDK: SSE streaming, session-gated tools, SQLite purchase-plan store
- Preserved conversation continuity across login, enabling shoppers to sign in mid-chat and continue with personalized help
- Expanded Madden Mobile commerce integration to expose player rosters, currencies, and progression data
- Shipped multi-item cart support to production in the shared commerce service behind the Madden Mobile Web Store
- Analyzed historical commerce data and comparable studies to project $157K in annual revenue lift from the assistant
- **Automation Tooling**: Deployed a shared-server agent fleet with per-user Docker workspaces and onboarded engineers
- Integrated GitLab and Jira context with agent-driven implementation, merge-request creation, and regression-test triggering, increasing team velocity without hurting reviewability, verifiability, and ownership of the code

Autodesk | Media & Entertainment May 2025 – Aug 2025
Software Engineering Intern – TypeScript, React, IndexedDB, Jenkins *Montreal, QC*

- Built an observability pipeline for live frame-by-frame review sessions in Creative Review's public beta
- Logged six event streams to IndexedDB: playback, pen strokes, text annotations, cursor position, rendering, session join/leave
- Built a visual debugger over the pipeline, aligning host and participant logs to pinpoint where clients diverged
- Resolved a production outage for two large studios within an afternoon by correcting feature-gating logic in the login flow

BETA Technologies | Stress & Structures Jan 2025 – Apr 2025
Software Engineering Intern – Python, PySide6, GitLab CI *Montreal, QC*

- Automated NASTRAN aircraft model preprocessing in Python, reducing a three-day manual workflow to under 10 seconds
- Developed a reusable Python library for NASTRAN model transformations: DMIG prep, numerical buckling, coordinate symmetry
- Implemented checkpoint-based rollback to generate model variants from shared state without re-parsing or copying

McGill AI Ethics Lab May 2024 – Dec 2024
Lead Researcher – Python, Flask, Go, React *Montreal, QC*

- Wrote a browser extension backend that parsed YouTube video transcripts around most-replayed segments to fact-check claims
- Integrated concurrent Go API clients into Python through CGo and ctypes, handling cross-language memory ownership
- Led prototype architecture and presented the research at McGill's undergraduate research symposium

Projects

Mini-C Compiler | Compilers, Java, JUnit, MIPS32

- Wrote an end-to-end compiler in Java for a C subset, targeting MIPS32 assembly
- Frontend: lexer, Pratt parser, AST, scoped name resolution, and type checking
- Backend: MIPS32 codegen for expressions, control flow, memory access, and function calls with a stack-based calling convention
- Implemented graph-coloring register allocation with backward liveness analysis, optimistic allocation, and spill-code generation
- Extended the language with classes, single inheritance, and virtual dispatch; built a JUnit/MARS test harness

Trading Fours | Recommender System, Python, Flask, React, XGBoost, Redis, MySQL, Docker, AWS

- Built a personalized Spotify recommender combining XGBoost genre predictions, listening history, and cosine-similarity ranking
- Deployed Flask and React containers to AWS EC2 through a GitHub Actions CI/CD pipeline pushing images to ECR
- Served the backend from preloaded Gunicorn workers sharing one model copy, with Redis session caching and MySQL pooling

Pi Talk | Agentic Tooling, TypeScript, Node.js, OpenAI TTS

- Published a text-to-speech extension for the Pi agent harness: streaming audio, pause/resume, live speed control
- Implemented Unicode-safe chunking, playback backpressure, timeouts, and cancellation when newer responses arrive

Skills

Languages: Java, Python, TypeScript, JavaScript, Go, C, C++, Swift, OCaml, SQL, Lua, Bash
Technologies: Linux, Git, Docker, AWS, GitHub Actions, Node.js, React, Flask, FastAPI, REST, Redis, PyTorch, scikit-learn, NGINX

Additional Projects

OS Kernel Simulator | Systems Programming, C, Make
- Created an OS simulator in C that loads shell scripts and tracks execution with process control blocks
- Implemented process scheduling with round-robin time slicing and priority aging for waiting processes
- Simulated demand paging with per-process page tables and LRU page replacement

Constrained Image Classification | Machine Learning, PyTorch, Bash, SLURM
- Developed CNN and feature-based odd-one-out image classifiers under a 25,000-parameter limit
- Built Bash/SLURM tooling to upload code, launch GPU training jobs, and retrieve results from a university cluster

Chani | Networking, OCaml, Lwt, TCP
- Wrote an OCaml TCP command-line chat app supporting concurrent client connections and server-side message broadcasting
- Implemented independent send/receive loops so clients could receive messages while awaiting input

Dictate | On-Device AI, Swift, AppKit, macOS, transcribe.cpp, Metal
- Built an offline macOS dictation app in Swift, replacing paid cloud dictation services like Superwhisper
- Implemented a global hotkey to record, on-device transcription with any local GGUF model, and pasting at the cursor
- Added transcript correction: a custom-word dictionary for misheard names plus configurable word or regex replacements

Lovebox | Raspberry Pi, JavaScript, Vite, Firebase, Python, Bash, systemd
- Built a Raspberry Pi touchscreen device, now in use for long-distance messaging, shared drawing, and Spotify playback
- Added touchscreen Wi-Fi setup and automated software updates with systemd, including rebuilds and selective service restarts